Project Title: Macro Trend Analysis & Cross-Domain Moderation on r/jobs

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1. Executive Summary & Problem Statement

The r/jobs subreddit (https://www.reddit.com/r/jobs/) functions as a vital online community for employment discussions, job hunting, and workplace grievances. However, labor market volatility creates sudden spikes in community anxiety, frequently triggering toxic discourse. This project establishes a dual-objective Machine Learning pipeline designed to perform macro-level trend tracking and micro-level content moderation without manually annotating Reddit training data; only a small annotated r/jobs test set is needed for evaluation. The two objectives are linked by tracking the rate of toxic comments within each discovered topic over time.

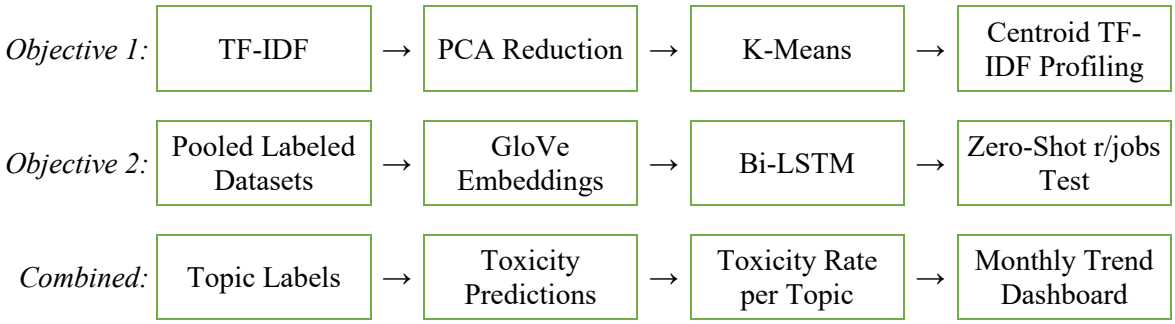


Figure 1. System pipeline architecture. Objective 1: Unsupervised Pipeline. Objective 2: Cross-Domain Transfer. Combined: Topic-Level Toxicity Trends.

2. Methodology & Machine Learning Techniques

Objective 1: Temporal Topic Discovery & Evolution (PCA & K-Means)

Goal: Extract underlying themes from monthly comment threads and track their volume shifts over time.

- *Feature Engineering.* Transform raw comment texts into numeric matrices using TF-IDF (Term Frequency–Inverse Document Frequency) vectorization to isolate unique, informative vocabulary.
- *Dimension Reduction.* Apply Principal Component Analysis (PCA) to compress the sparse high-dimensional vocabulary matrix down to a dense vector space ($d = 50$) while retaining maximum global variance.
- *Clustering & Validation.* Execute K-Means Clustering on the reduced space. Optimal cluster count (k) is determined using the Elbow Method and Silhouette Scores.
- *Interpretable Centroid Profiling.* To resolve PCA opacity, post-processing calculates the mean TF-IDF scores for documents within each cluster. Extracting the top-weighted terms yields clean, human-readable labels (e.g., “interview anxiety”, “layoff notices”) to chart month-over-month trend evolution.

Business Use Case: For the r/jobs moderation team, the monthly topic trends show which discussions are growing before they overwhelm the subreddit. When a theme such as layoff notices starts rising, moderators can schedule extra coverage, tighten AutoModerator filters, and consolidate repetitive posts into a pinned megathread. Rising themes also show where the wiki, FAQ, and sidebar resources need updating (e.g., severance rights, interview preparation), and whether new rules or post flairs are needed, such as for scam job postings. Combined with the toxicity rate per topic, the team can focus review effort on the themes where discourse is turning hostile, and check in the following month whether an intervention reduced toxicity in that cluster. The dashboard's early-warning value is checked by comparing the volume of the layoff-related cluster against a public layoff tracker (e.g., layoffs.fyi) over the same months.

Objective 2: Cross-Domain Moderation & Data Viability Assessment (Bi-LSTM)

Goal: Train a toxicity classifier on external pre-labeled comments and evaluate out-of-domain generalization to Reddit, testing whether platform-independent data can replace manual annotation of Reddit training data.

Core Research Question: Can toxicity patterns learned from pre-labeled datasets pooled across other platforms (e.g., YouTube comments, Wikipedia talk pages, news-site comments, online game chat) be used to classify r/jobs comments for content moderation, without manual labeling of Reddit training data?

- *Motivation & Strategy:* Manual annotation of specialized forum posts is time-consuming and labor-intensive. This objective assesses if pre-annotated datasets from rich public environments (YouTube comments, discussion-forum and news-site comments, game chat) suffice to train high-performing classifiers for Reddit.
- *Training Data:* Pre-labeled toxicity datasets from four sources are pooled into a single training set. Because each source defines toxicity differently, labels are mapped to a common binary scheme (toxic / non-toxic) before pooling:
  - YouTube comments: *YouTube Toxicity Data* (https://www.kaggle.com/datasets/reihanenamdari/youtube-toxicity-data); comments labeled toxic via the IsToxic flag.
  - Wikipedia talk-page discussions: *Jigsaw Toxic Comment Classification Challenge* (https://www.kaggle.com/competitions/jigsaw-toxic-comment-classification-challenge); comments labeled toxic if any of the six toxicity categories is flagged.
  - News-site comment sections: *Jigsaw Unintended Bias in Toxicity Classification (Civil Comments)* (https://www.kaggle.com/competitions/jigsaw-unintended-bias-in-toxicity-classification); comments labeled toxic if the annotator toxicity score is 0.5 or higher.
  - Online game chat: *GOSU.ai English Dota 2 Chat* (https://www.kaggle.com/datasets/danielfesalbon/gosu-ai-english-dota-chat); messages labeled toxic if scored as mild or strong toxicity.
- *Model Architecture:* Construct a Bidirectional Long Short-Term Memory (Bi-LSTM) neural network using pre-trained GloVe word embeddings.
- *Architectural Justification.* Unlike standard unidirectional LSTMs that process text left-to-right, a Bi-LSTM runs two parallel layers (forward and backward). This allows the network to process complete sentential context simultaneously - crucial for capturing context-dependent insults, sarcasm, and implicit toxic tone.
- *Cross-Domain Evaluation:* The network is trained exclusively on the pooled external data and tested zero-shot on a small, manually annotated test subset of r/jobs comments, labeled with the same binary scheme. Transfer performance is evaluated using Precision, Recall, Macro F1-Score, and PR-AUC.

Integration: Topic-Level Toxicity Trends

The two objectives are combined by applying the trained classifier to the r/jobs comments clustered in Objective 1. For each topic cluster, the share of comments predicted as toxic is tracked month over month, showing which themes (e.g., layoff notices) drive spikes in toxic discourse.